UNIVERSITY OF EDINBURGH

COLLEGE OF SCIENCE AND ENGINEERING

SCHOOL OF INFORMATICS

INFR09027 COMPUTER COMMUNICATIONS AND NETWORKS

Friday 3rd May 2013

14:30 to 16:30

INSTRUCTIONS TO CANDIDATES

Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY BE USED IN THIS EXAMINATION

Year 3 Courses

Convener: K. Kalorkoti
External Examiners: K. Eder, T. Field

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. (a) Give a balanced analysis of protocol layering, outlining the advantages and disadvantages. *[4 marks]*

(b) A communication protocol has an `n`-layer hierarchy. Applications generate messages of length `M`, and an `h`-byte header is added at each layer. What fraction of the network bandwidth is filled with headers? *[2 marks]*

(c) Describe the two broad approaches towards the implementation of congestion control in computer networks? *[4 marks]*

(d) Consider a packet-switched network over which is transmitted data of size `z` bits as a series of packets over a `k`-hop path. Each packet contains `d` data bits and `h` header bits, where the combined size of the header and data bits, `(d + h)`, in each packet is much smaller than the overall size of the transmitted data `(z)`. What is the value of `d` which minimises the total delay, assuming that the bit rates of the lines are uniformly `b` bits/sec (you may assume that the propagation delay is negligible)? *[7 marks]*

(e) Consider distributing a file of size `F` equal to 10 Gbytes to `N` peers. The server has an upload rate of $u_s = 50$ Mbps, and each peer has a download rate of $d_i = 2$ Mbps and an upload rate of u. For `N` $= 10$ and 100, and $u =$ 250 Kbps and 500 Kbps, tabulate the minimum distribution times for each of the combination of `N` and u for both client-server distribution and P2P distribution. *[8 marks]*

2. (a) Describe the Ethernet Media Access Control (MAC) protocol. [*3 marks*]

   (b) When a file is transferred between two computers, two acknowledgement strategies are possible. In the first one, the file is chopped up into packets, which are individually acknowledged by the receiver, but the file transfer as a whole is not acknowledged. In the second one, the packets are not acknowledged individually, but the entire file is acknowledged when it arrives. Discuss the relative merits of these two approaches. [*4 marks*]

   (c) Suppose a host has a 1 MByte file that is to be sent to another host over a 1 Mbps link with 10 ms one-way propagation delay. The file takes 1 second of CPU time to compress 50%, or 2 seconds to compress 60% (ignore queueing delays).

      i. Calculate the bandwidth at which the combined time taken for compression and transmission for each of the compression options equals that of the uncompressed case. [*4 marks*]

      ii. Does propagation delay also get affected? Explain either why, or, why not? [*2 marks*]

   (d) Suppose a large organization with 1000 routers employs a simple 2-level routing hierarchy with the routers equally distributed between 20 areas (as in OSPF) to keep the routing table size small. Assume that, unlike OSPF, every router needs to keep an entry in its routing table indicating the route to reach areas other than the one it belongs to. At what network size in terms of the total number of routers will it no longer be possible to keep the routing table size at or below 100 entries with the above mentioned 2-level hierarchy? [*7 marks*]

   (e) Give five reasons why an application developer might choose UDP over TCP as the underlying transport protocol. [*5 marks*]

3. (a) Describe the services offered to the immediate layers above in a protocol by each of the following: TCP, UDP, IP and Ethernet. [*2 marks*]

(b) BGP and RIP are examples of Internet routing protocols that rely on TCP or UDP for their message exchanges. Since they are conceptually part of the network layer, why do they not just use raw IP packets? [*3 marks*]

(c) Both Address Resolution Protocol (ARP) and Domain Name System (DNS) rely on caches; ARP cache entry lifetimes are typically 10 minutes, while DNS cache lifetimes are on the order of days. Justify this difference. What undesirable consequences might there be in having too long a DNS cache entry lifetime? [*6 marks*]

(d) Two CSMA/CD stations are each trying to transmit long (multiframe) files. After each frame is sent, they contend for the channel, using the binary exponential backoff algorithm. What is the probability that the contention ends on round k, and what is the mean number of rounds per contention period? [*6 marks*]

(e) Suppose your organization decides to upgrade the existing Ethernet from 10 Mbps to 100 Mbps without changing the network configuration, node locations and cabling.

i. What would you as a network designer do to maintain the same efficiency as before? [*3 marks*]

ii. Suppose that you are also tasked with improving the efficiency of the network after the upgrade. What would you recommend your users to do? Discuss your approach. [*5 marks*]